

Exercise 3: Google App Engine SDK and Web App Development

Aim

To install the Google App Engine SDK and develop a Hello World / simple web application using Python.

Procedure

Step 1: Install Google Cloud CLI & Python

Ensure Python 3 is installed on the system.

```
ubuntu@ubuntu:~$ sudo apt install python3
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 3
```

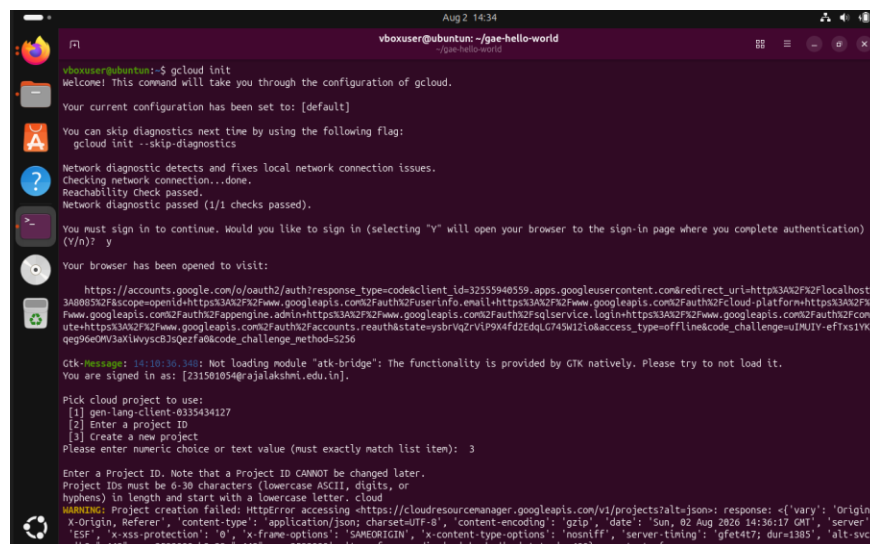
Download and install the Google Cloud CLI (contains the modern App Engine SDK) from cloud.google.com/sdk/docs/install:

```
curl -O https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-cloud-cli-  
linux-x86_64.tar.gz  
tar -xf google-cloud-cli-linux-x86_64.tar.gz  
./google-cloud-sdk/install.sh
```

Step 2: Initialize GCloud & Install Components

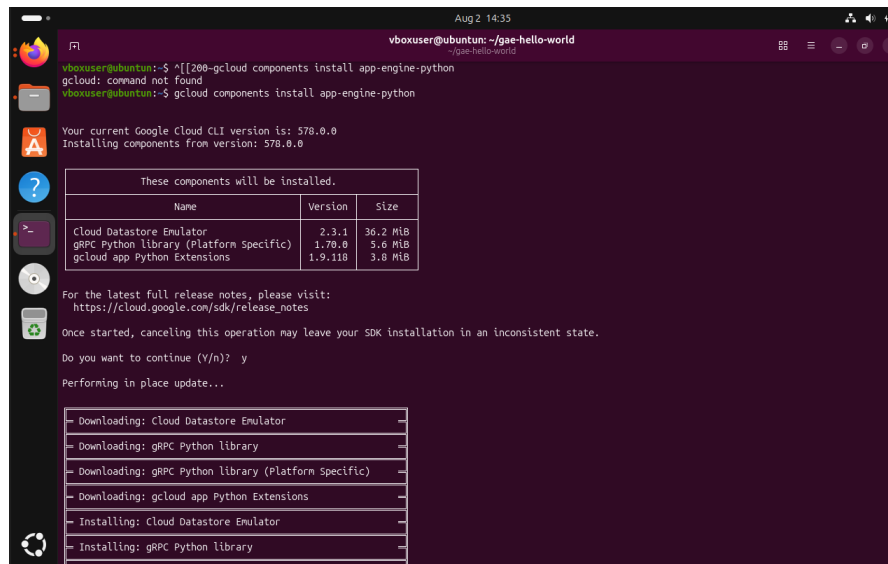
```
gcloud init
```

Sign in to your Google account when prompted.



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gcloud components install app-engine-python

A terminal window titled 'vboxuser@ubuntu: ~/gae-hello-world' showing the command 'gcloud components install app-engine-python' being executed. The terminal output indicates the current Google Cloud CLI version is 578.0.0 and shows the components to be installed in a table.

These components will be installed.		
Name	Version	Size
Cloud Datastore Emulator	2.3.1	36.2 MiB
gRPC Python library (Platform Specific)	1.78.0	5.6 MiB
gcloud app Python Extensions	1.9.118	3.8 MiB

The terminal also shows a prompt to continue (Y/n) and a progress bar for downloading and installing the components.

Step 3: Create the App Directory

```
mkdir gae-hello-world  
cd gae-hello-world
```

Step 4: Create Application Code (main.py)

```
from flask import Flask  
app = Flask(__name__)  
  
@app.route("/")  
def hello():  
    return "Hello World from Google App Engine!"  
  
if __name__ == "__main__":  
    app.run(host="127.0.0.1", port=8080, debug=True)
```

Step 5: Create Configuration File (app.yaml)

```
runtime: python39  
handlers:  
- url: /.*  
  script: auto
```

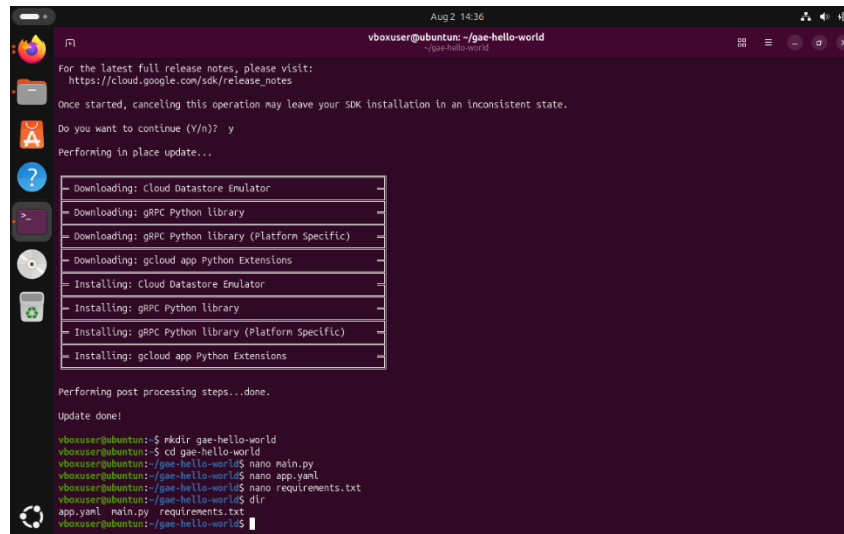
Step 6: Create requirements.txt

```
Flask==2.0.1
```

Output

Files created in project directory (main.py, app.yaml, requirements.txt):

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The image shows a terminal window titled 'vboxuser@ubuntu: ~/gae-hello-world' with a timestamp of 'Aug 2 14:36'. The terminal displays the following text:

```
For the latest full release notes, please visit:
https://cloud.google.com/sdk/release_notes

Once started, cancelling this operation may leave your SDK installation in an inconsistent state.
Do you want to continue (Y/n)? y
Performing in place update...

- Downloading: Cloud Datastore Emulator
- Downloading: gRPC Python library
- Downloading: gRPC Python library (Platform Specific)
- Downloading: gcloud app Python Extensions
- Installing: Cloud Datastore Emulator
- Installing: gRPC Python library
- Installing: gRPC Python library (Platform Specific)
- Installing: gcloud app Python Extensions

Performing post processing steps...done.
Update done!

vboxuser@ubuntu:~/gae-hello-world$ mkdir gae-hello-world
vboxuser@ubuntu:~/gae-hello-world$ cd gae-hello-world
vboxuser@ubuntu:~/gae-hello-world$ nano main.py
vboxuser@ubuntu:~/gae-hello-world$ nano app.yaml
vboxuser@ubuntu:~/gae-hello-world$ nano requirements.txt
vboxuser@ubuntu:~/gae-hello-world$ dir
app.yaml  main.py  requirements.txt
vboxuser@ubuntu:~/gae-hello-world$
```

Result

Thus, the Google App Engine SDK was installed and a simple Hello World web application was developed using Python and Flask.